
CSE (DATA SCIENCE)

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Title of the project:

AI-Driven E-Commerce Recommendation System with Digital Twin Simulation

Project Batch No: A11

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Abstract:

E-commerce platforms have become an important part of modern shopping, where customers expect personalized product recommendations and fast service. Most existing recommendation systems use purchase history, browsing history, ratings, and reviews to suggest products. Although these systems improve the shopping experience, they still face several challenges, including limited personalization, dependence on historical data, poor real-time adaptation, lack of recommendation transparency, and reduced performance during periods of high server traffic. These limitations affect both customer satisfaction and overall system efficiency.

To address these challenges, this project proposes an **Adaptive E-Commerce Recommendation System Using Big Data Analytics, Digital Twin Simulation, and AI-Based Traffic Optimization**. The system analyzes customer engagement factors such as product views, clicks, searches, cart activities, purchases, ratings, and reviews to better understand customer preferences. **Big Data Analytics** is used to process large volumes of customer data efficiently, while a **Multi-Agent AI** framework generates personalized recommendations based on user behavior.

The proposed system also integrates **Digital Twin Simulation** to virtually evaluate recommendation strategies before they are presented to users, improving recommendation quality and reliability. **Explainable AI (XAI)** provides users with clear reasons for each recommendation, increasing transparency and trust. A **Self-Learning Feedback Engine** continuously updates the recommendation model based on user interactions, while the **AI-Based Traffic Optimization** module monitors server load and efficiently manages recommendation requests during peak shopping periods to ensure smooth system performance.

The proposed framework improves recommendation accuracy, scalability, transparency, response time, and overall customer experience while maintaining efficient resource utilization. **Future improvements** may include integrating **voice-based shopping assistants, Augmented Reality (AR) product visualization, blockchain-based review verification, and federated learning** to further enhance personalization, security, privacy, and the overall shopping experience.

**Project Coordinator
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Project